

Advanced Linear Algebra: Foundations to Frontiers

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2.3.2 Geometric interpretation ¶

02.3.2 A geometric interpretation of the SVD, part 1

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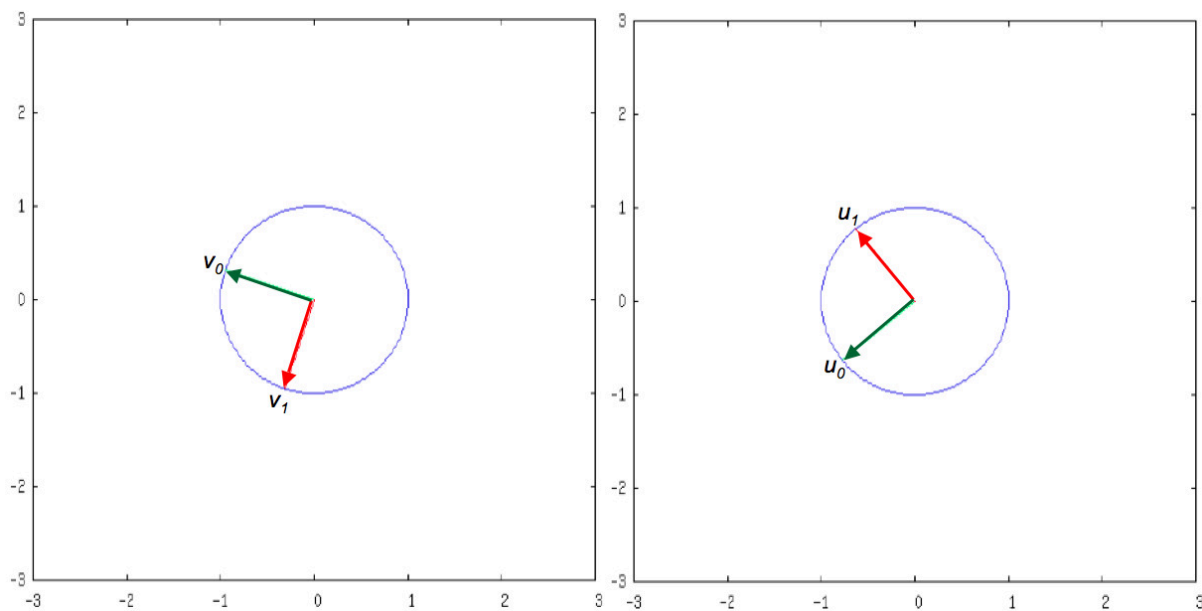
We will now illustrate what the SVD Theorem tells us about matrix-vector multiplication (linear transformations) by examining the case where $A \in \mathbb{R}^{2 \times 2}$. Let $A = U\Sigma V^T$ be its SVD. (Notice that all matrices are now real valued, and hence $V^H = V^T$.) Partition

$$A = \left(\begin{array}{c|c} u_0 & u_1 \end{array} \right) \left(\begin{array}{c|c} \sigma_0 & 0 \\ \hline 0 & \sigma_1 \end{array} \right) \left(\begin{array}{c|c} v_0 & v_1 \end{array} \right)^T.$$

Since U and V are unitary matrices, $\{u_0, u_1\}$ and $\{v_0, v_1\}$ form orthonormal bases for the range and domain of A , respectively:

$\mathbb{R}^2$: Domain of A :

$\mathbb{R}^2$: Range (codomain) of A :



Let us manipulate the decomposition a little:

$$\begin{aligned}
 A &= \left(\begin{array}{c|c} u_0 & u_1 \end{array} \right) \left(\begin{array}{c|c} \sigma_0 & 0 \\ \hline 0 & \sigma_1 \end{array} \right) \left(\begin{array}{c|c} v_0 & v_1 \end{array} \right)^T \\
 &= \left[\left(\begin{array}{c|c} u_0 & u_1 \end{array} \right) \left(\begin{array}{c|c} \sigma_0 & 0 \\ \hline 0 & \sigma_1 \end{array} \right) \right] \left(\begin{array}{c|c} v_0 & v_1 \end{array} \right)^T \\
 &= \left(\begin{array}{c|c} \sigma_0 u_0 & \sigma_1 u_1 \end{array} \right) \left(\begin{array}{c|c} v_0 & v_1 \end{array} \right)^T.
 \end{aligned}$$

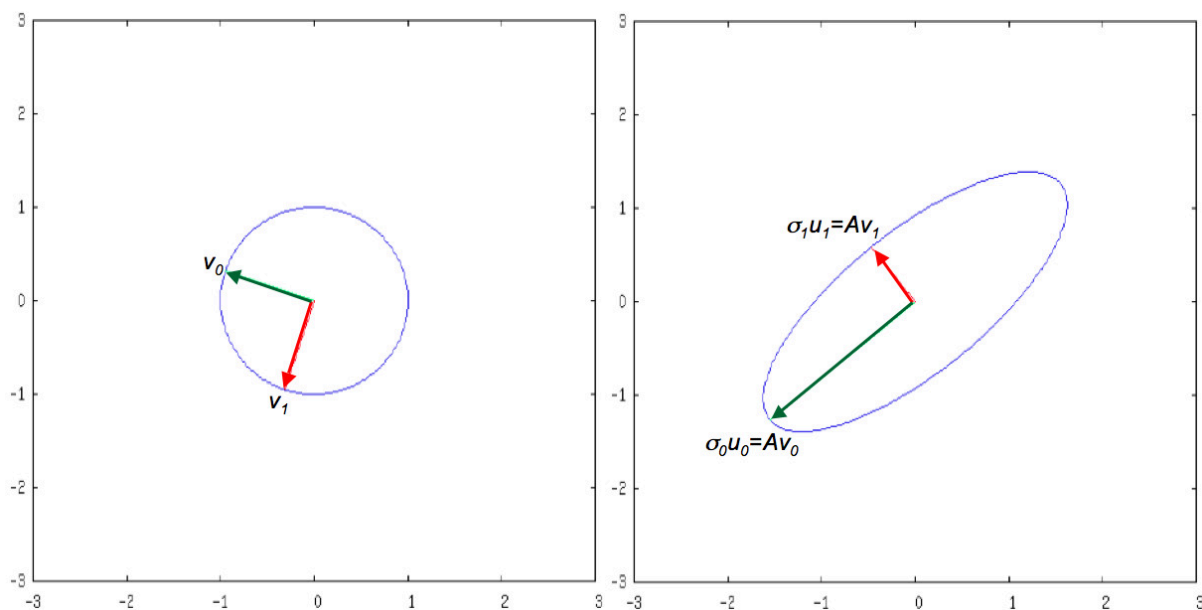
Now let us look at how A transforms v_0 and v_1 :

$$Av_0 = \left(\begin{array}{c|c} \sigma_0 u_0 & \sigma_1 u_1 \end{array} \right) \left(\begin{array}{c|c} v_0 & v_1 \end{array} \right)^T v_0 = \left(\begin{array}{c|c} \sigma_0 u_0 & \sigma_1 u_1 \end{array} \right) \left(\begin{array}{c} 1 \\ 0 \end{array} \right) = \sigma_0 u_0$$

and similarly $Av_1 = \sigma_1 u_1$. This motivates the pictures in [Figure 2.3.2.1](#).

$\mathbb{R}^2$: Domain of A :

$\mathbb{R}^2$: Range (codomain) of A :



$\mathbb{R}^2$: Domain of A :

$\mathbb{R}^2$: Range (codomain) of A :

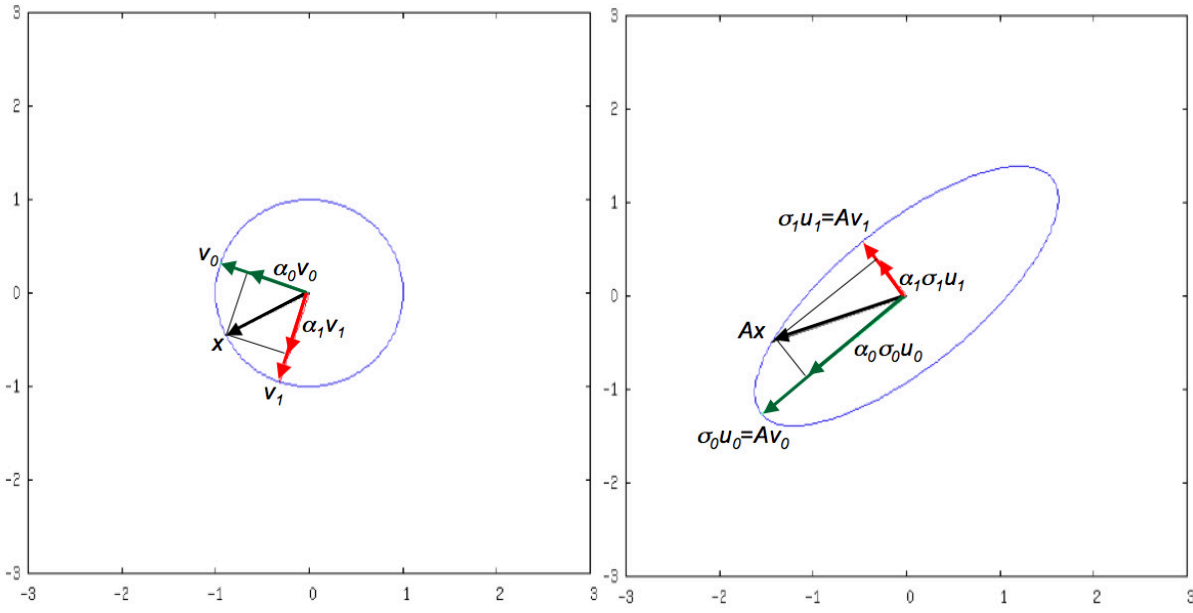


Figure 2.3.2.1. Illustration of how orthonormal vectors v_0 and v_1 are transformed by matrix $A = U\Sigma V$.

Next, let us look at how A transforms any vector with (Euclidean) unit length.

Notice that $x = \begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix}$ means that

$$x = \chi_0 e_0 + \chi_1 e_1,$$

where e_0 and e_1 are the unit basis vectors. Thus, χ_0 and χ_1 are the coefficients when x is expressed using e_0 and e_1 as basis. However, we can also express x in the basis given by v_0 and v_1 :

$$\begin{aligned} x &= \underbrace{VV^T}_I x = \begin{pmatrix} v_0 & v_1 \end{pmatrix} \begin{pmatrix} v_0 & v_1 \end{pmatrix}^T x = \begin{pmatrix} v_0 & v_1 \end{pmatrix} \begin{pmatrix} \frac{v_0^T x}{v_1^T x} \end{pmatrix} \\ &= \underbrace{v_0^T x}_{\alpha_0} v_0 + \underbrace{v_1^T x}_{\alpha_1} v_1 = \alpha_0 v_0 + \alpha_1 v_1 = \begin{pmatrix} v_0 & v_1 \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix}. \end{aligned}$$

Thus, in the basis formed by v_0 and v_1 , its coefficients are α_0 and α_1 . Now,

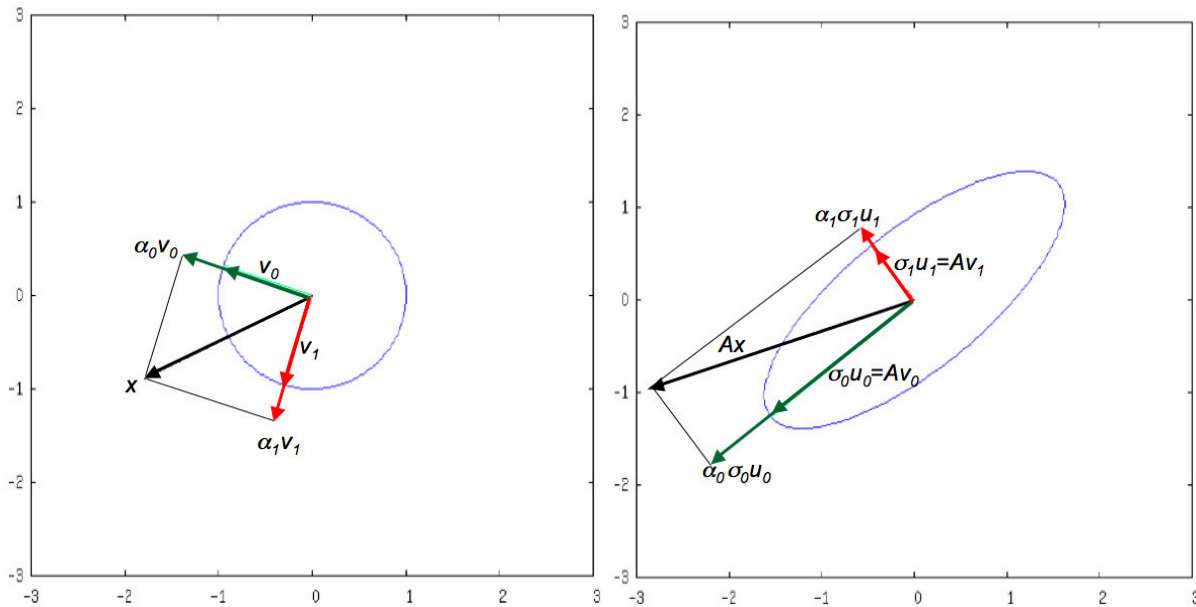
$$\begin{aligned} Ax &= \begin{pmatrix} \sigma_0 u_0 & \sigma_1 u_1 \end{pmatrix} \begin{pmatrix} v_0 & v_1 \end{pmatrix}^T x \\ &= \begin{pmatrix} \sigma_0 u_0 & \sigma_1 u_1 \end{pmatrix} \begin{pmatrix} v_0 & v_1 \end{pmatrix}^T \begin{pmatrix} v_0 & v_1 \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix} \\ &= \begin{pmatrix} \sigma_0 u_0 & \sigma_1 u_1 \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix} = \alpha_0 \sigma_0 u_0 + \alpha_1 \sigma_1 u_1. \end{aligned}$$

This is illustrated by the following picture, which also captures the fact that the unit ball is mapped to an oval with major axis equal to $\sigma_0 = \|A\|_2$ and minor axis equal to σ_1 , as illustrated in [Figure 2.3.2.1](#) (bottom).

Finally, we show the same insights for general vector x (not necessarily of unit length):

$\mathbb{R}^2$: Domain of A :

$\mathbb{R}^2$: Range (codomain) of A :



- Another observation is that if one picks the right basis for the domain and codomain, then the computation Ax simplifies to a matrix multiplication with a diagonal matrix. Let us again illustrate this for nonsingular $A \in \mathbb{R}^{2 \times 2}$ with

$$A = \underbrace{\begin{pmatrix} u_0 & u_1 \end{pmatrix}}_U \underbrace{\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_1 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} v_0 & v_1 \end{pmatrix}^T}_V.$$

- Now, if we chose to express y using u_0 and u_1 as the basis and express x using v_0 and v_1 as the basis, then

$$\begin{aligned} \underbrace{UU^T}_I y &= \underbrace{UU^T}_{\hat{y}} y = (u_0^T y)u_0 + (u_1^T y)u_1 \\ &= \begin{pmatrix} u_0 & u_1 \end{pmatrix} \begin{pmatrix} \frac{u_0^T y}{u_1^T y} \end{pmatrix} = U \underbrace{\begin{pmatrix} \hat{\psi}_0 \\ \hat{\psi}_1 \end{pmatrix}}_{\hat{y}} \\ \underbrace{VV^T}_I x &= \underbrace{VV^T}_{\hat{x}} x = (v_0^T x)v_0 + (v_1^T x)v_1 \\ &= \begin{pmatrix} v_0 & v_1 \end{pmatrix} \begin{pmatrix} \frac{v_0^T x}{v_1^T x} \end{pmatrix} = V \underbrace{\begin{pmatrix} \hat{\chi}_0 \\ \hat{\chi}_1 \end{pmatrix}}_{\hat{x}}. \end{aligned}$$

- If $y = Ax$ then

$$\underbrace{UU^T}_{\hat{y}} y = \underbrace{U\Sigma V^T}_{Ax} x = U\Sigma \hat{x}$$

- so that

$$\hat{y} = \Sigma \hat{x}$$

and

$$\begin{pmatrix} \hat{\psi}_0 \\ \hat{\psi}_1 \end{pmatrix} = \begin{pmatrix} \frac{\sigma_0 \hat{\chi}_0}{\sigma_1 \hat{\chi}_1} \end{pmatrix}.$$

Remark 2.3.2.2. The above discussion shows that if one transforms the input vector x and output vector y into the right bases, then the computation $y := Ax$ can be computed with a diagonal matrix instead: $\hat{y} := \Sigma \hat{x}$. Also, solving $Ax = y$ for x can be computed by multiplying with the inverse of the diagonal matrix: $\hat{x} := \Sigma^{-1} \hat{y}$.

These observations generalize to $A \in \mathbb{C}^{m \times n}$: If

$$y = Ax$$

then

$$U^H y = U^H A \underbrace{V V^H}_I x$$

so that

$$\underbrace{U^H y}_{\hat{y}} = \Sigma \underbrace{V^H x}_{\hat{x}}$$

(Σ is a rectangular "diagonal" matrix.)

02.3.2 A geometric interpretation of the SVD, part 2

fit width

